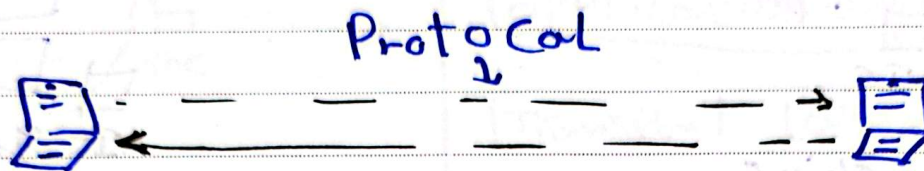


Network Stack

1. introduction to Network

→ Network definition: Exchange Data with another external Machine

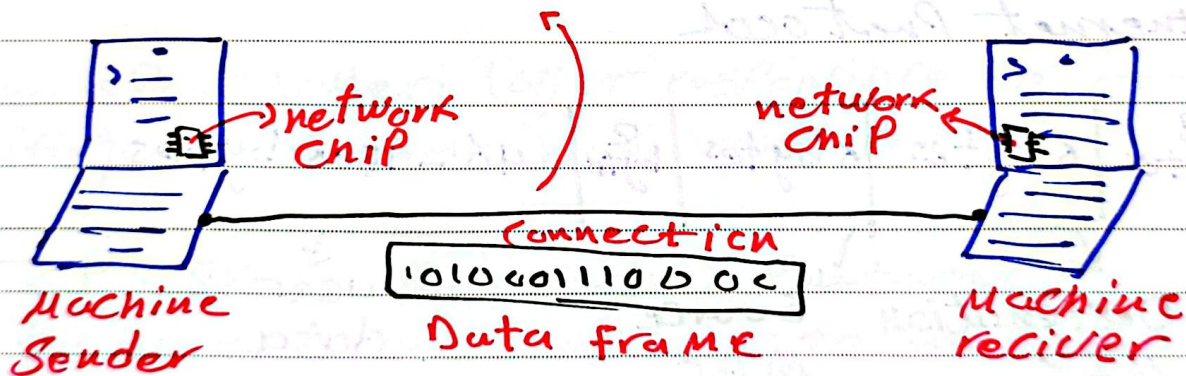


Protocol: the way of exchanging data can receiver and sender understand

→ Data called Frame

→ network Main component :-

Direct / Indirect Connection



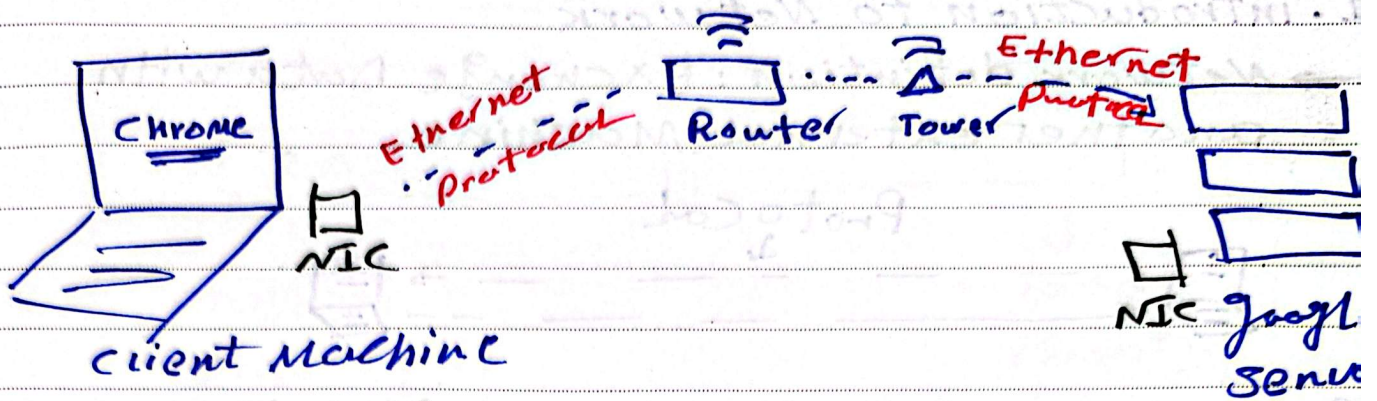
network chip:- this chip can able device to deal with network without Network chip can't connect

Direct Connection: A ↔ B

Indirect Connection: A ↗ C ↘ B

A can connect to B through C

→ internet network component:-

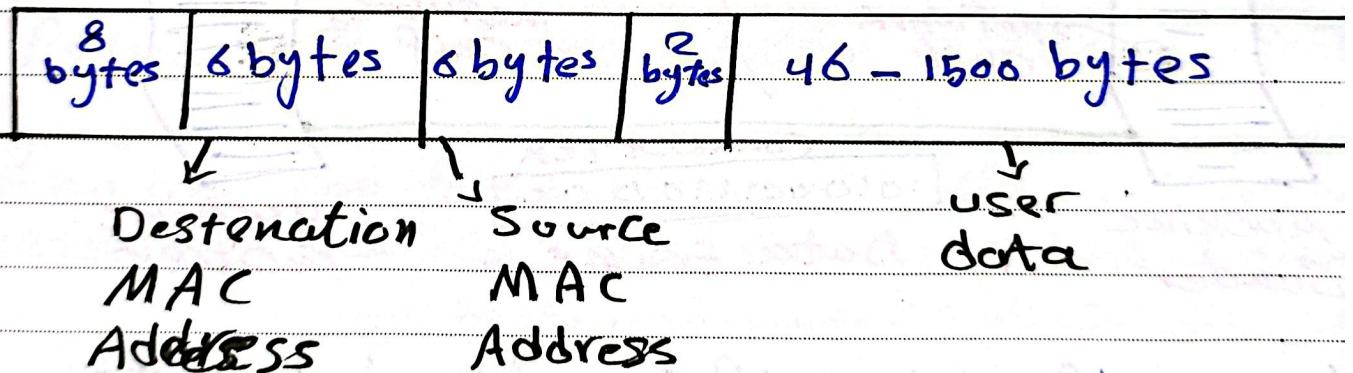


→ Function of each component

• Router

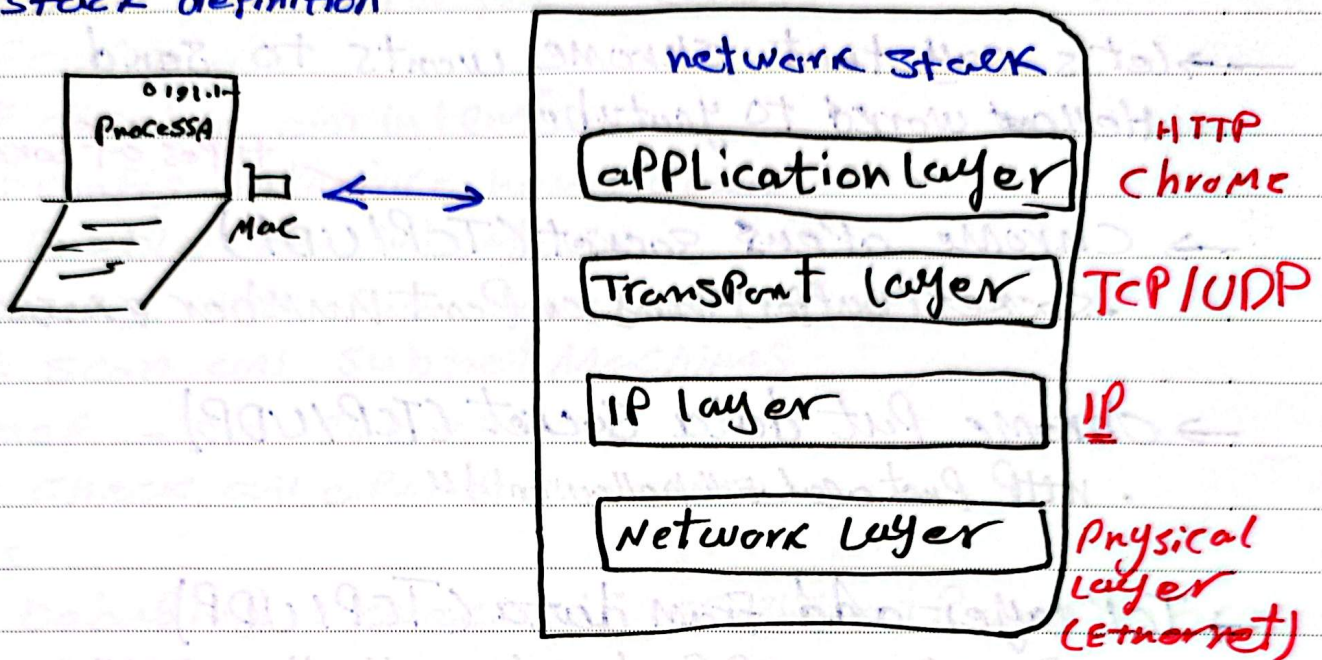
- Set IPs local network
- Capable to access Public network

• Ethernet Protocol



Network Stack

1. stack definition



Application layer: Layer of application like Chrome using HTTP Protocol

Transport layer: layer responsible for transmitting (data) frame over TCP/UDP

TCP: Transmission Control Protocol

the task of this Protocol is to carry packets across the internet and ensure the successful delivery of messages and data across network

Secure

UDP: user datagram Protocol

send without verifying its delivery
faster

example:-

→ let's say that Chrome wants to send Hello world to youtube

types of sockets

→ Chrome opens socket (TCP/UDP)

- socket(buffer) has a Port number ex: 5050

→ Chrome put data socket (TCP/UDP)

- http Protocol + "helloworld"

→ TCP layer add fram data (TCP/UDP)

- TCP/UDP + http Protocol + "helloworld"

→ IP layer add IP data

- IP Source + destination + TCP/UDP + http + "hello..."

→ add mac address and send frame

- From IP to "helloworld" puts in user data in ethernet Protocol

- then fill ethernet data: MAC destination and Source Mac address

2. Commands:-

- # Show all interfaces conds
`ifconfig -a`
- # disable an interface
`ifconfig interface-name down`
- # Enable an interface
`ifconfig interface-name up`
- # Scan all Subnet Machines
`nmap -sn Subnet/24`
- # Check all opened sockets
`ss`
- # Debugging : check if specific IP can be accessed
`ping <IP>`
- # Access remote Machine using SSH
`ssh`

3. exercise

- # require nent's
- # 1. activate network chip.
- # 2. Scan all Machines connected in your subnet
- # 3. Ping on of this Machine to see if there is a path.
- # 4. access one of the Machines using ssh
- # 5. Check sockets opened through ssh
- # 6. Monitor Frames with Machine using Wireshark